

💡 What is Dynamic Routing?

When part of the URL changes dynamically (like an ID or username) and the page shows different data accordingly, that's called Dynamic Routing.

Example:

You visit:

- /user/saharsh
- /user/rahul
- /user/123

And based on URL, different data render hoti hai.

simple words → Dynamic routing ka matlab hota hai ki **route ka kuch hissa dynamic ho**, jo URL ke through change hota hai.

/user/1

/user/2

/user/saharsh

Ye sab ek common route pattern follow karte hain: /user/:id

Yahan :id ek dynamic parameter hai jo runtime pe change hota hai.

```
import React from "react";
import { Routes, Route, BrowserRouter } from "react-router-dom";
import User from "../pages/User";

function App() {
  return (
    <div>
      <BrowserRouter>
        <Routes>
          <Route path="/user/:id" element={<User />} />
        </Routes>
      </BrowserRouter>
    </div>
  );
}

export default App;
```

```
import React from 'react'
import { useParams } from 'react-router-dom'

const User = () => {
  const {id} = useParams();
  return (
    <div>
      <h3>Welcome User : {id}</h3>
    </div>
  )
}

export default User
```

2. Optional Parameters (Segment)

segment matlab do / ke bich ka → /user/:id → yahape user ek segment hai id bhi hai

React Router **v6.4+** se tum :id? jaisa optional parameter use kar sakte ho.

✓ Syntax:

```
<Route path="/user/:id?" element={<User />} />
```

Yahan:

- /user bhi valid hai
- /user/123 bhi valid hai

```
import React from "react";
import { Routes, Route, BrowserRouter } from "react-router-dom";
import User from "../pages/User";
import Home from "../pages/Home";

function App() {
  return (
    <div>
      <BrowserRouter>
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/user/:id?" element={<User />} />
        </Routes>
      </BrowserRouter>
    </div>
  );
}

export default App;
```

3. <NavLink> kya hai?

<NavLink> is just like <Link>, but with **extra feature**:

✔ It allows us to **style the active link** (jo current page pe ho).

```
<NavLink to="/about" style={({ isActive }) => ({ color: isActive ? 'red' : 'black', fontWeight: isActive ? 'bold' : 'normal' })} > About </NavLink>
```

Feature	Description
<NavLink>	Like <Link>, but adds active styling
isActive	Built-in flag to check current active
className / style	Functions to apply conditional styles

npm i -D tailwindcss postcss autoprefixer
isme -d ka matlab ky hai

Iska matlab kya hai?

- --save-dev ya -D ka use un packages ke liye hota hai jo **development ke dauraan** chahiye hote hain, **na ki production mein**.
- Ye packages aapke package.json ke "devDependencies" section mein add kiye jaate hain.

API

what is API ?

- application programming interface
- we need data from DB for project
- but js can't connect with database because the js is client side scripting language and DB is server side
- so we make api in other language like java, php or node , etc
- aur wo jo data hota hai wo ek json data hota hai

API methods

- get
- post
- put/patch
- delete

```
import React, { useEffect, useState } from 'react';
import './API.css'; // CSS in separate file (optional but clean)

const API = () => {
  const [users, setUsers] = useState([]);

  useEffect(() => {
    getUserData();
  }, []);

  const getUserData = async () => {
    const url = 'https://dummyjson.com/users';
    try {
      const response = await fetch(url);
      const data = await response.json();
      setUsers(data.users);
    } catch (err) {
      console.error("API Error:", err);
    }
  };
};
```

install json server make api